

Mehak Sharma

Software Developer — IT Consultant — Cloud & DevOps — Applied AI & ML

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Professional Profile

Analytical and solution-driven **Software Developer and M.Sc. AI & Data Science** graduate with hands-on experience designing, developing, and optimizing distributed cloud systems and AI/ML integrations. Proven track record in translating complex business and customer requirements into robust, scalable software architectures. Experienced in building automated CI/CD and testing pipelines, implementing closed-loop quality monitoring, and orchestrating enterprise LLM systems. Enthusiastic about tackling challenging technical problems within agile, client-focused consulting teams.

Technical Skills

Software Engineering: Python, TypeScript, SQL, FastAPI, Pydantic, REST APIs, Microservices, Object-Oriented Design
Cloud, DevOps & Automation: AWS (SageMaker, Glue, S3, RDS), Docker, GitLab CI/CD, Automated Testing, Cloud Architecture

AI & Machine Learning Integration: LLM integration, OpenAI Agents SDK, LangChain, RAG architectures, MLflow, PySpark, vector search, model evaluation

Quality & Testing: Test-driven automation, output validation, closed-loop monitoring, regression test suites, security-conscious API design

Consulting & Methods: Requirements analysis, Agile/Scrum methodologies, cross-functional collaboration, technical documentation

Professional Experience

AI Engineer (Working Student) — Allianz Technology SE

Munich, Germany

Mar 2025 – Present

- Analyzed internal stakeholder requirements to design and implement an enterprise-grade **multi-agent software system** using OpenAI Agents SDK, autonomously routing service requests across enterprise applications.
- Built and integrated robust **backend toolchains and REST APIs** connecting Azure OpenAI with legacy enterprise systems via FastMCP, FastAPI, and LangChain RAG pipelines.
- Implemented comprehensive **automated validation and test mechanisms**, including structured output checks, regression benchmarks, and real-time KPI monitoring to detect drift and maintain operational reliability.
- Enforced security-conscious API patterns, structured logging, and automated error handling across cloud-hosted microservices.
- Collaborated in an agile setting with cross-functional technical and domain teams to iterate rapidly on customer requirements and deploy scalable prototypes.

Data Engineer — L&T Technology Services

Bengaluru, India

Sep 2022 – Mar 2024

- Architected and optimized large-scale Spark data pipelines on AWS (Glue, SageMaker, S3, RDS) for enterprise client Delta Airlines, increasing processing reliability by **35%**.
- Automated model deployment workflows and tracking utilizing MLflow and Docker, establishing standardized release pipelines across distributed teams.
- Delivered reusable software components, data governance frameworks, and lineage tracking systems to ensure consistent high data quality for downstream analytics.
- Participated in Agile sprint planning, technical backlog grooming, and cross-team architectural reviews.

Associate Engineer — L&T Technology Services

Mysuru, India

Feb 2022 – Aug 2022

- Designed and maintained automated **GitLab CI/CD pipelines** and automated testing suites for computer vision applications, cutting manual deployment overhead.
- Partnered with senior engineers to implement unit and integration test environments within an Agile development lifecycle.

Key Projects

Enterprise Agentic AI Platform — Allianz Technology SE

2025

- Architected a modular enterprise software platform integrating multi-agent orchestration, RAG pipelines, automated output validation, and KPI-based monitoring for real-time process automation.

Automated Industrial Code Generation & Analysis System

Mar 2024 – Sep 2024

- Designed and prototyped a retrieval-augmented software tool (LangChain, GPT-4, CodeLlama) for automated code generation, syntax validation, and comprehension in industrial computing environments.

Education

M.Sc. Artificial Intelligence & Data Science — Deggendorf Institute of Technology, Germany

Mar 2024 – Mar 2027

B.Tech Computer Science & Engineering — Punjab Technical University, India

Apr 2018 – Apr 2022

Languages & Communication

English: C1 (Fluent, Professional Working Proficiency) — **German:** A2 (Actively Learning) — **Hindi:** Native